

Learning Objectives

- Explain how missing values, outliers, and duplicate data are handled when cleaning a dataset.
- Describe data type conversion, normalization and standardization, and the reshaping of data to fit a purpose.

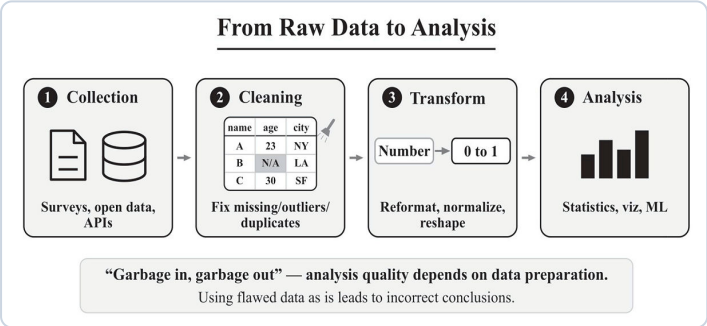
Real data arrives messy: blank cells, a value ten times too large, the same row entered twice, dates written three different ways. Before any analysis can be trusted, the data must be cleaned and transformed to fit the question. There is a rule in computing — "garbage in, garbage out" — which means careful calculation on flawed data still gives a flawed answer. This lesson is about turning raw data into data you can actually rely on.

? Guiding question: Why must data be cleaned and transformed before analysis, and what choices does each fix involve?

Point!

1. The Data Cleaning and Transformation Stages

(1) Collected data often cannot be used for analysis as is. Adjusting issues such as missing values, outliers, and duplicate data, and arranging the data into a format that fits the purpose of the analysis, is called **(data cleaning)** and **(data transformation)**.



From raw data to analysis: collect, clean, transform, analyze

(2) If issues are left unaddressed during analysis, incorrect conclusions may be drawn. For this reason, cleaning and transformation are important stages that support the accuracy and reliability of the analysis.

2. Handling Missing Values

(1) Using data containing missing values (values where data is absent) directly in analysis can prevent calculations from being performed correctly or cause biased results. For missing values, one of the following three responses is chosen.

Key Fact:
Raw data is rarely ready to use. Cleaning fixes missing values, outliers, and duplicates; transformation converts types, rescales, and reshapes data to fit the purpose. "Garbage in, garbage out" — the quality of any analysis depends on this preparation.

Key Concepts:

- data cleaning / transformation
- missing values, outliers, duplicates
- data type conversion
- normalization / standardization
- reshaping (pivot, join, aggregation)

KEY TERMS
data cleaning — fixing missing values, outliers, duplicates.
data transformation — reshaping/rescaling data to fit the purpose.

PAUSE & THINK

A blank cell in a survey — delete the row, fill it in, or flag it? What decides?



Fix blanks and judge outliers before analysis.

Response	Description	When to apply
Deletion	Remove rows or columns containing missing values from the data	When there are few missing values and the impact on the whole is small
Imputation	Fill in the missing parts with values such as the mean, median, or surrounding values	When missing values are scattered without bias, and replacing them with representative values from other data does not significantly change the overall trend
Flagging	Leave a marker indicating that the value was missing, and analyze the data as is	When the missingness itself has meaning (e.g., the fact that someone refused to answer)

(2) Which response to choose is determined by the proportion and pattern of missing values, and the purpose of the analysis. For example, imputing the mean for an item where missing values account for a large proportion of all data may produce values that are far from reality. There is no single correct response; judgment according to the purpose is required.

3. Handling Outliers and Duplicate Data

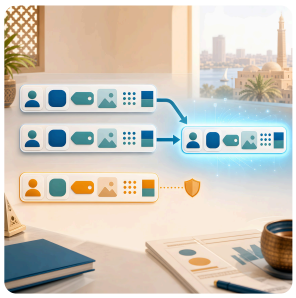
(1) Outliers (values that deviate significantly from other data) are not necessarily errors. They may be **(abnormal values)** caused by input or measurement errors, or they may be values that reflect reality. *e.g.,* If the sales data of a certain store shows that one day has remarkably high sales, it may be an input error, or it may genuinely reflect a sale or event on that day.

KEY TERMS
missing value — a value that is absent.
outlier — a value far from the others.
abnormal value — an outlier caused by an error.

(2) Handling outliers begins with identifying the cause.

① When the cause is an input or measurement error…correct or delete the value

② When it is a value that reflects reality…in principle, keep it and decide how to handle it based on the purpose of the analysis



Remove only genuinely duplicated records.

(3) Duplicate data (a state in which the same content exists in multiple records) affects the results of aggregation and is, in principle, removed. However, it is important to confirm that the data is genuinely duplicated. Even if records appear identical, they may include data that should be retained as separate records. *e.g.,* If the same customer purchases the same product twice on the same day, the records are separate transactions; combining them into one would make the sales aggregation incorrect.

4. Data Type Conversion

(1) Collected data is sometimes difficult to handle in its original format. Arranging the format to make data usable for analysis is called **(data type conversion)**. Typical examples include the following.

Type of conversion	Description	Example
Conversion from string to numeric	Convert numbers stored as strings into numeric values that can be used in calculations	“123” → 123
Unification of date format	Align dates with different notations into a single format	“2026/4/1,” “2026-04-01,” “01 Apr 2026” → 2026-04-01
Unification of notation	Combine values with the same meaning into a single notation	A mix of “Male/Female” and “M/F” → standardized to “Male/Female”

(2) If formats are not aligned, data cannot be aggregated correctly. For example, if numbers are stored as strings, sums and means cannot be calculated; if date notations are not unified, data from the same day will be treated as belonging to different days. For this reason, data type conversion is a step that must be checked before analysis.

5. Normalization and Standardization

(1) When the units or ranges (scales) of data differ greatly between variables, variables with larger values may strongly influence the analysis results. To prevent this, processing is performed to align the scale of data. Typical methods are **(normalization)** and **(standardization)**.

Method	Description	Main purpose
Normalization	Convert data to a range from 0 to 1	Align data with different units into a comparable form
Standardization	Convert data to a form with mean 0 and standard deviation 1	Position values relative to the spread of the data

(2) For example, when analyzing variables with different units and scales together, such as height (cm) and weight (kg), normalization can align both into the range from 0 to 1, making them easier to compare. Specifically, if height data ranges from 160 cm to 180 cm, normalizing converts 160 cm → 0, 170 cm → 0.5, 180 cm → 1, so the minimum becomes 0 and the maximum becomes 1.

(3) Standardization aligns data based on how far each value is from the mean. For example, when test scores from multiple subjects are standardized, even if the mean and spread differ between subjects, the position of each student can be compared.

6. Reshaping Data According to Purpose

(1) Processing that changes the arrangement or grouping of data according to the purpose of the analysis is called **(data reshaping)**. The following three operations are typical.

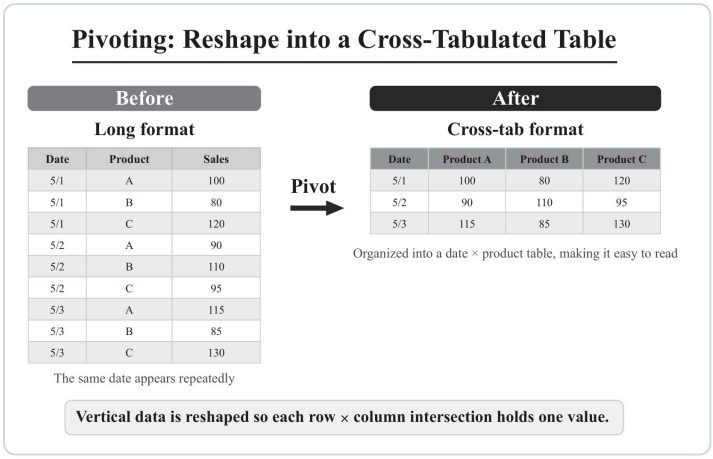
PAUSE & THINK

Height in cm and weight in kg — why might the bigger numbers dominate an analysis?



Normalization rescales variables to a common range.

Operation	Description	Example
(Pivot)	Reshape vertically arranged data into a table of row × column combinations	Reshape data with date, product, and sales arranged vertically into a “date × product” table
(Join)	Connect multiple datasets via a common item	Connect a student roster and a grade sheet by “student ID” into a single table
(Aggregation)	Calculate sums, means, etc. for each group and summarize	Compute monthly totals from daily sales



Pivoting: reshaping long-format data into a cross-tabulated table

(2) Which operation to perform depends on the purpose of the analysis. For example, if you want to see “monthly sales trends,” you need to aggregate daily sales data by month. It is important to clarify the purpose of the analysis first, and then plan how to reshape the data.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Data cleaning is work performed before data collection.

B The responses to missing values are deletion, imputation, and flagging—three in total.

C All outliers are caused by input or measurement errors, so they must always be deleted.

D Standardization is processing that converts data to a form with mean 0 and standard deviation 1.

(2) Match each description (a–c) with the most appropriate processing from the choices below (A–C).

a Conversion of data to a range from 0 to 1

b Processing that connects different datasets via a common item

c Processing that calculates sums, means, etc. for each group and summarizes

💡 PAUSE & THINK

Is a store's one huge sales day always an error — or could it be real?

[Choices] **A** Aggregation **B** Join **C** Normalization

✓ Solution

(1)

A Data cleaning is the work of arranging collected data, and is performed after collection. Therefore, ×.

B Correct as the responses to missing values. Therefore, ○.

C Outliers include both abnormal values caused by input errors and values that reflect reality. They are not always deleted. Therefore, ×.

D Correct definition of standardization. Therefore, ○.

(2)

a: Conversion to 0–1 is normalization. **C**

b: Connecting via a common item is join. **B**

c: Summarizing by group is aggregation. **A**

Try

1 Answer the following questions.

1. What is the term for the processing that fills in the missing parts of data with values such as the mean or median?
2. What is the term for the processing that converts data to a range from 0 to 1?
3. What is the term for the processing that converts data to a form with mean 0 and standard deviation 1?
4. What is the term for the operation that connects multiple datasets via a common item?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate response to missing values.

A When there are few missing values and the impact on the whole is small, delete rows containing missing values.

B When missing values are scattered randomly, fill in the missing parts with values such as the mean.

C When the missingness itself has meaning, leave a marker indicating that the value was missing and analyze the data.

D Impute the mean and analyze data on items where most of the data is missing.

EXAM-STYLE QUESTION

Explain, using "garbage in, garbage out," why data preparation matters as much as the analysis itself. **[6] Refer to:** one cleaning fix and one transformation.

(2) For each of situations (a, b), choose the most appropriate data processing from the choices below (A, B).

a You want to align date data with mixed notations such as “2026/4/1,” “2026-04-01,” and “01 Apr 2026.”

b You want to compare variables with different units and scales together, such as height (cm) and weight (kg).

[Choices] **A** Data type conversion (standardization of date format)
B Normalization

(3) From the following options (A–D), choose the one that most appropriately describes the handling of outliers.

A Outliers are all errors in the data, so delete all of them as soon as they are found.

B For outliers, identify the cause first; correct or delete them if they are due to input errors, and in principle keep them if they reflect reality.

C Outliers do not affect analysis, so no special handling is needed.

D Outliers can always be resolved by replacing them with the mean.

Think as an Engineer — Investigate & Decide

Your class collected a survey on daily screen time, but the raw table is messy: some cells are blank, one entry reads “9999” hours, dates are written three different ways, and two rows look identical. You prepare the data for analysis.

☐ **Analyze the raw table as is** ☐ **Clean and transform it first**

1 · Inspect the data. List each problem you can see find; for each, say which category it is (missing value, outlier, duplicate, or format issue) and note in one phrase how it would distort the analysis if left unfixed.

2 · Decide a fix and its risk. For the blank cells, choose deletion, imputation, or flagging and justify it from the proportion of blanks. For “9999,” say how you would check its cause — a typo, a wrong unit (minutes entered as hours), or a code entered for a refused answer — before removing it. Name one respondent whose true answer a careless fix might erase.

3 · Plan the transformation. State the one format conversion you must run so the dates aggregate correctly, and whether you would normalize the screen-time column. **In pairs,** compare cleaning plans and agree on which single fix carries the greatest risk of distorting the results — and why.

Give evidence for your answer.

PAUSE & THINK

If most of a column is blank, does filling it with the mean help — or invent data?

KEY TERMS

pivot — long rows → row × column table.
join — link datasets by a common item.
aggregation — summarize by group.

Stuck? Handle one problem type at a time in order: unify formats first, then check the outlier's cause, then decide the blanks, then remove only true duplicates — and never delete a value before you know why it is there.

🌍 In a New Context

A shop's sales table lists each day's sales for products A, B, and C on separate rows, but the manager wants one row per day with a column per product. Name the reshaping operation needed, and name the operation instead required to get each product's monthly total.

❓ Lesson Question — Answered

The lesson asked why data must be cleaned and transformed before analysis and what choices each fix involves. Collected data is rarely usable as is, and flawed data yields flawed conclusions no matter how careful the later calculation — garbage in, garbage out. Cleaning handles missing values (by deletion, imputation, or flagging, chosen from their proportion and meaning), outliers (judged by cause: correct or delete errors, keep real values), and duplicates (removed only when genuinely duplicated). Transformation makes data comparable and analyzable: data type conversion unifies formats (string to numeric, date formats, notation), normalization rescales to 0–1 and standardization to mean 0 and standard deviation 1 so no variable dominates by size, and reshaping — pivot, join, aggregation — arranges data to fit the question. Each step is a judgment guided by the purpose, not a fixed rule.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

If collected data is used directly in analysis, incorrect conclusions may be drawn. For this reason, (a), which addresses issues, and (b), which arranges the format to fit the purpose of the analysis, are necessary. For missing values, an appropriate response is chosen from deletion, (c), and flagging, depending on the proportion of missing values and the purpose of the analysis. Also, when

⭐ REFLECT & CHALLENGE

Reflect: Why can “filling in” missing data sometimes mislead more than leaving it out? **Challenge:** You have one survey column where 70% of answers are blank. Argue whether to impute, flag, or drop the column, and state the one condition under which your choice would change.

units or ranges differ greatly between variables, the scale is aligned through (d), which converts data to a range from 0 to 1, or (e), which converts data to a form with mean 0 and standard deviation 1.

1. Fill in the blanks (a)–(e).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that corresponds to data type conversion.

A Filling in missing values with the mean.

B Standardizing notations such as “2026/4/1” and “01 Apr 2026” into “2026-04-01.”

C Aligning the scales of height and weight to between 0 and 1.

D Aggregating daily sales into monthly totals.

(2) From the following options (A–D), choose the one that most appropriately describes normalization and standardization.

A Normalization is processing that converts data to a form with mean 0 and standard deviation 1.

B Standardization is processing that converts data to a range from 0 to 1.

C Normalization is used to align data with different units into a comparable form.

D Standardization is used to enlarge the scale of variables.

(3) For each processing (a–c), indicate which corresponds to pivot, join, or aggregation.

a Connecting a student roster and a grade sheet by “student ID” into a single table.

b Calculating monthly totals from daily sales data.

c Reshaping daily product sales into the form of a “date × product” table.

[Choices] **A** Pivot **B** Join **C** Aggregation

⭐ Key takeaway

Remember:
Clean first (missing values, outliers, duplicates), then transform (convert types, normalize or standardize, reshape by pivot, join, aggregation). Every fix is a judgment set by the purpose — garbage in, garbage out.